PAPER 1

1a. Three examples of Executive Support Systems/Executive Information Systems are:

- Dashboards: These systems provide executives with a visual representation of key performance indicators and critical business metrics, allowing them to monitor the overall health of the organization and make informed decisions.

- Strategic Planning Systems: These systems help executives in long-term planning and strategic decision-making by providing them with relevant data, market trends, and forecasting capabilities.

- Balanced Scorecard Systems: These systems enable executives to track and manage the performance of the organization across multiple dimensions, such as financial, customer satisfaction, internal processes, and learning and growth.

b. Three examples of Management Information Systems are:

- Sales Management Systems: These systems collect and analyze sales data, including revenue, customer information, and sales trends, to assist managers in making sales-related decisions and optimizing sales strategies.

- Inventory Management Systems: These systems track inventory levels, monitor stock movements, and generate reports on inventory status, helping managers ensure optimal stock levels and streamline supply chain processes.

- Human Resources Information Systems: These systems manage employee information, such as payroll, attendance, performance evaluations, and training records, to support HR managers in various tasks, including recruitment, compensation, and performance management.

c. Three examples of Knowledge Management Systems are:

- Knowledge Repositories: These systems store and organize explicit knowledge, such as documents, reports, and best practices, allowing employees to access and share knowledge across the organization.

- Collaboration Platforms: These systems facilitate communication, collaboration, and knowledge sharing among employees, providing features like shared workspaces, discussion forums, and wikis.

- Expert Systems: These systems leverage artificial intelligence and expert knowledge to provide specialized advice and solutions in specific domains, assisting employees in complex decision-making or problem-solving tasks.

d. Three examples of Transaction Processing Systems are:

- Point of Sale (POS) Systems: These systems capture and process sales transactions at retail locations, including scanning products, calculating totals, and updating inventory and financial records.

- Online Payment Systems: These systems handle online payment transactions, securely processing credit card payments, electronic fund transfers, and other forms of online financial transactions.

- Reservation Systems: These systems manage and process reservation requests for services like flights, hotels, and car rentals, ensuring accurate bookings, availability tracking, and payment processing.

e. Three examples of Office Automation Systems are:

- Email Systems: These systems enable electronic communication and exchange of messages within the organization and with external parties.

- Document Management Systems: These systems facilitate the creation, storage, retrieval, and version control of digital documents, improving collaboration and document workflow.

- Calendar and Scheduling Systems: These systems help employees manage their schedules, appointments, and meetings, ensuring efficient time management and coordination.

Question Two:

1. Four distinct parts of Information Technology systems in an organization are:

- Hardware: This includes the physical components of IT systems, such as computers, servers, network devices, and storage devices.

- Software: These are the programs and applications that run on the hardware and provide specific functionalities, including operating systems, productivity software, and specialized business applications.

- Networks: Networks are the infrastructure that connects computers, devices, and systems within an organization, enabling data communication and resource sharing.

- Data: Data represents the information stored and processed by IT systems, including databases, files, and documents.

2. Four impacts of Information Technology in an organization are:

- Improved Efficiency: Information Technology automates tasks, streamlines processes, and enhances productivity, allowing organizations to accomplish more in less time.

- Enhanced Communication and Collaboration: IT systems enable fast and effective communication, both within the organization and with external stakeholders, facilitating collaboration and knowledge sharing.

- Access to Information: Information Technology provides instant access to vast amounts of information, enabling organizations to make data-driven decisions and respond quickly to changing

market conditions.

- Competitive Advantage: Effective utilization of IT systems can give organizations a competitive edge by enabling innovation, improving customer experiences, and optimizing operations.

3. The difference between Information Systems and Information Technology is:

Information Systems refer to the broader concept of managing and leveraging information within an organization. It encompasses the people, processes, and technologies involved in collecting, storing, processing, and disseminating information to support business operations and decision-making.

Information Technology, on the other hand, specifically refers to the technology infrastructure and tools used to manage information. It includes hardware, software, networks, and data management systems that enable the processing, storage, and transmission of information.

4. Example of how Information Systems can be used: An example of how Information Systems can be used is in customer relationship management (CRM). CRM systems capture and analyze customer data, including interactions, preferences, and purchase history, to enhance customer engagement, improve sales and marketing strategies, and drive customer satisfaction and loyalty.

5. Four primary issues in Computer Ethics are:

- Privacy: Concerns related to the protection and appropriate use of personal and sensitive information.

- Intellectual Property: Ethical considerations surrounding copyrights, patents, and intellectual property rights.

- Cybersecurity: The ethical implications of hacking, unauthorized access, and malicious activities that compromise the security and integrity of computer systems.

- Social Impact: Ethical dilemmas arising from the use of technology, such as the impact on employment, social inequality, and privacy in the digital age.

Question Three:

1. Data Communication Terminologies:

i. Channel: In data communication, a channel refers to the physical medium or path through which data is transmitted, such as copper wires, fiber optic cables, or wireless radio waves.

ii. Bandwidth: Bandwidth is the capacity or data transfer rate of a communication channel, indicating the amount of data that can be transmitted in a given time. It is typically measured in bits per second (bps) and represents the speed of data transmission.

2. A computer network is a collection of interconnected devices, such as computers, servers, routers, switches, and other networking equipment, that are linked together to facilitate communication, data sharing, and resource sharing.

3. Two principle benefits of computer networks are:

- Resource Sharing: Networks allow multiple users to access and share resources, such as printers, storage devices, and internet connections, improving efficiency and reducing costs.

- Communication and Collaboration: Networks enable users to communicate and collaborate through email, instant messaging, video conferencing, and shared file systems, fostering teamwork and information exchange.

4. Three classifications of computer networks are:

- Local Area Network (LAN): A LAN covers a limited geographical area, such as a home, office building, or campus. It connects devices within a close proximity and typically operates at high speeds.

- Wide Area Network (WAN): A WAN spans large distances and connects multiple LANs or other networks together, often using public or private telecommunication links. Examples include the internet and private leased lines.

- Metropolitan Area Network (MAN): A MAN covers a larger area than a LAN but smaller than a WAN, typically serving a city or a metropolitan region. It provides high-speed connectivity for organizations within the same geographical area.

5. Network Topology refers to the physical or logical arrangement of devices and

connections in a computer network. It defines how devices are interconnected and how data flows between them.

6. The difference between Logical and Physical topologies:

- Logical Topology: Logical topology defines the flow of data in a network conceptually, regardless of the physical layout. It describes the logical connections and addressing schemes used to transmit data between devices.

- Physical Topology: Physical topology refers to the actual physical layout of devices, cables, and other network components. It defines the physical arrangement and connections between devices in a network.

7.Three types of topologies in computer networks are:

- Bus Topology: In a bus topology, all devices are connected to a central cable (bus) in a linear manner. Data transmitted by one device is received by all other devices on the bus.

- Star Topology: In a star topology, all devices are connected to a central hub or switch. Data transmitted by one device is routed through the hub/switch to the intended destination device.

- Ring Topology: In a ring topology, devices are connected in a closed loop, with data flowing in one direction. Each device acts as a repeater to transmit the data to the next device in the ring.

PAPER 2

Question One:

1. Four distinct parts of Information Technology systems in an organization are:

a. Hardware: This refers to the physical components of the IT system, including computers, servers, routers, switches, and other devices used to process and store data.

b. Software: This encompasses the programs, applications, and operating systems

that run on the hardware and enable users to perform various tasks. Examples include word processing software, database management systems, and customer relationship management (CRM) software.

c. Networks: Networks connect different devices within an organization, allowing them to communicate and share information. This includes local area networks (LANs), wide area networks (WANs), and the internet.

d. Data: Data is a crucial component of IT systems. It refers to the information that is stored, processed, and transmitted within the organization. This includes databases, files, documents, and any other form of digital information.

2. Impacts of Information Technology in an organization:

a. Increased Efficiency: Information Technology automates manual tasks, streamlines processes, and improves overall efficiency. It enables organizations to accomplish tasks faster and with fewer errors, leading to increased productivity.

b. Enhanced Communication: IT systems facilitate communication within an organization, allowing employees to collaborate and share information easily. Email, instant messaging, video conferencing, and other communication tools enable real-time communication across different locations.

c. Improved Decision Making: IT systems provide access to accurate and timely information, enabling better decision making at all levels of an organization. Data analytics and business intelligence tools help in analyzing data and extracting valuable insights.

d. Competitive Advantage: Information Technology can give organizations a competitive edge by enabling innovative products and services, improving customer experiences, and allowing for efficient business processes. It helps organizations adapt to changing market dynamics and stay ahead of the competition.

3. The difference between Information Systems and Information Technology:

Information Systems (IS) refer to the broader framework that encompasses the management and use of information within an organization. It involves the processes, people, and technologies used to collect, organize, store, process, and

disseminate information to support decision making and achieve organizational goals.

Information Technology (IT) is a subset of Information Systems and focuses specifically on the technology infrastructure and tools used to manage information. IT deals with the hardware, software, networks, and data involved in the storage, retrieval, and transmission of information.

Data Flow Examples:

a. Simplex Mode: In simplex mode, data can only flow in one direction. A common example is a television broadcast where the information is transmitted from the broadcasting station to the television viewers.

b. Half Duplex: In half duplex mode, data can flow in both directions, but not simultaneously. Walkie-talkies are an example of a half-duplex communication system where one party can transmit while the other party listens, and vice versa.

c. Full Duplex: In full duplex mode, data can flow simultaneously in both directions. A telephone conversation is an example of a full-duplex communication system, where both parties can speak and listen at the same time.

d. Asynchronous: In asynchronous data flow, data is transmitted in irregular intervals with the sender and receiver using start and stop bits to synchronize communication. An example is the transmission of data through email.

e. Synchronous: In synchronous data flow, data is transmitted in a continuous stream without start and stop bits. Examples include real-time video conferencing or live streaming where data is transmitted in real-time without interruption.

Four components of data communications system:

1. Sender: The sender is the source of information or data that needs to be transmitted. It can be a computer, a user, a sensor, or any other device capable of generating or capturing data.

2. Receiver: The receiver is the destination of the transmitted data. It can

be another computer, a display device, a storage device, or any other device that can receive and process the data.

3. Medium: The medium refers to the physical channel or pathway through which the data is transmitted. It can be wired, such as copper or fiber-optic cables, or wireless, such as radio waves or satellite links.

4. Protocols: Protocols are a set of rules and conventions that govern the formatting, transmission, and interpretation of data. They define how data is packaged, addressed, error-checked, and delivered between sender and receiver. Protocols ensure reliable and efficient communication between devices. Examples include TCP/IP for internet communication or Ethernet for local area networks.

PAPER 3

Question One:

a) Three categories of input devices:

1. Keyboards: Keyboards are a common input device that allows users to input text and commands into a computer. They consist of a set of keys, including alphanumeric keys, function keys, and special keys for specific tasks.

2. Pointing Devices: Pointing devices are used to control the movement of the cursor or pointer on the screen. Examples include a mouse, trackball, touchpad, or touch screen. They allow users to interact with graphical user interfaces (GUI) and select objects or options.

3. Scanners: Scanners are input devices used to convert physical documents or images into digital format. They capture the content of the document or image and create a digital representation that can be stored or manipulated on a computer. Examples include flatbed scanners, document scanners, or barcode scanners.

b) Reasons why the use of biometric devices is encouraged in security-sensitive systems:

1. Increased Security: Biometric devices use unique physiological or behavioral characteristics of individuals, such as fingerprints, iris patterns, or voiceprints, for identification and authentication. These characteristics are difficult to forge or replicate, making biometrics more secure than traditional password-based systems.

2. Non-Transferable and Non-Shared: Biometric traits are tied to individuals and cannot be transferred or shared like passwords or access cards. This reduces the risk of unauthorized access or identity theft, as biometric data is unique to each individual.

c) Importance of considering ergonomics before buying hardware:

1. User Comfort and Health: Ergonomics focuses on designing hardware to optimize user comfort and reduce the risk of physical discomfort or injuries. Considering ergonomics before purchasing hardware helps ensure that users can work comfortably and minimize the risk of repetitive strain injuries or musculoskeletal disorders.

2. Productivity and Efficiency: Ergonomic hardware design can enhance productivity and efficiency. Ergonomic keyboards, mice, and chairs, for example, are designed to reduce fatigue and strain, allowing users to work for longer periods without discomfort. This can lead to improved productivity and overall performance.

d) Eight input devices:

1. Keyboard 2. Mouse 3. Scanner 4. Webcam

5. Microphone 6. Touchscreen 7. Joystick 8. Gamepad

e) Eight output devices:

1. Monitor/Display 2. Printer 3. Speakers 4. Headphones

5. Projector 6. Plotter 7. Actuators (e.g., motors or lights)

8. Haptic devices (e.g., force-feedback controllers)

f) Impact of IT on various aspects:

a. Health: IT has revolutionized the healthcare industry, enabling electronic health records, telemedicine, and remote patient monitoring. It has improved access to medical information, facilitated faster diagnoses, and enhanced patient care and safety.

b. Money: IT has transformed financial systems, enabling online banking,

electronic transactions, and digital currencies. It has made financial services more accessible, efficient, and secure, allowing for faster and convenient transactions.

c. Education: IT has revolutionized education by providing access to online learning platforms, digital educational resources, and collaborative tools. It has expanded learning opportunities, personalized education, and facilitated distance learning.

d. Entertainment: IT has had a significant impact on the entertainment industry. Streaming services, online gaming, social media platforms, and digital content creation have transformed how we consume and interact with entertainment. It has provided new avenues for creativity, distribution, and engagement.

PAPER 4

QUESTION ONE

(a) Network Topology: Network topology refers to the physical or logical arrangement of computers, devices, and communication links in a computer network. It defines how the network nodes are connected and how data flows within the network.

(b) The two main parts of a computer are:

1. Hardware: Hardware refers to the physical components of a computer system that can be touched and interacted with. This includes components such as the central processing unit (CPU), memory modules, storage devices, input devices (e.g., keyboard, mouse), output devices (e.g., monitor, printer), and networking devices (e.g., routers, switches).

2. Software: Software refers to the programs, instructions, and data that are stored electronically on a computer. It includes the operating system, application software, and data files. Software enables the hardware components to perform specific tasks and operations.

(c) Distinguishing the following:

(i) Storage and Memory:

- Storage: Storage refers to long-term data storage, typically on non-volatile media such as hard disk drives (HDDs), solid-state drives (SSDs), or optical disks. It retains data even when the power is turned off and is used to store files, programs, and operating systems.

- Memory: Memory, also known as RAM (Random Access Memory), is temporary storage that holds data and instructions that the CPU needs to process in real-time. It is volatile, meaning it loses its contents when the power is turned off. Memory is used to store actively used programs and data for faster access by the CPU.

(ii) Data and Information:

- Data: Data refers to raw, unprocessed facts, symbols, or statistics. It is a representation of facts, numbers, or characters without any context or meaning.

- Information: Information is processed and organized data that provides meaning and context. It is the result of analyzing, interpreting, and structuring data to make it meaningful and useful for decision-making or understanding.

(iii) Synchronous transmission and asynchronous transmission:

- Synchronous transmission: In synchronous transmission, data is sent in a continuous stream at a fixed rate and synchronized timing. Both the sender and receiver use a common clock to coordinate the transmission. It is commonly used for high-speed, reliable data transfer, such as in telecommunications and networking.

- Asynchronous transmission: In asynchronous transmission, data is sent in individual units with start and stop bits to indicate the beginning and end of each unit. The timing between each unit can vary, and there is no fixed rate. It is commonly used for slower-speed data transfer, such as serial communication between devices.

(iv) Field and record:

- Field: In a database or spreadsheet, a field is a single piece of data, representing a specific attribute or characteristic of an entity. For example, in a database of

employees, a field could be "name," "age," or "salary."

- Record: A record is a collection of related fields that form a complete set of information about a specific entity or item. It represents a single instance or row in a database table. Using the example of an employee database, a record would contain fields like name, age, salary, and other relevant information for a particular employee.

(v) Data rate and bandwidth:

- Data rate: Data rate refers to the speed at which data is transmitted or processed over a network or computer system. It is measured in bits per second (bps) or its multiples, such as kilobits per second (Kbps) or megabits per second (Mbps).

- Bandwidth: Bandwidth refers to the maximum amount of data that can be transmitted over a network or communication channel in a given time. It is usually measured in bits per second (bps) or its multiples (Kbps, Mbps, etc.). It represents the capacity or throughput of the channel.

(d) Input devices:

1. Keyboard: A keyboard is an input device that allows users to input text, commands, and other characters into a computer system by pressing individual keys.

2. Mouse: A mouse is a pointing device that allows users to control the movement of the cursor on the screen and select objects by clicking buttons.

3. Scanner: A scanner is a device that captures images or text from physical documents and converts them into digital formats that can be stored and processed on a computer.

4. Microphone: A microphone is an input device used to capture audio or voice input. It converts sound waves into electrical signals that can be processed and recorded by the computer.

5. Webcam: A webcam is a video camera that captures live video and streams it in real-time over the Internet or to a computer system.

Output devices:

1. Monitor: A monitor, also known as a display or screen, is an output device that visually presents the processed data, images, and videos to the user.

2. Printer: A printer is an output device that produces hard copies of documents, images, or other digital content on paper or other media.

3. Speakers: Speakers are output devices that produce audio output, allowing users to hear sounds, music, and other audio content from the computer.

4. Projector: A projector is an output device that displays images or video onto a larger screen or surface, often used in presentations or home theater systems.

5. Headphones: Headphones are output devices worn over the ears that deliver audio output directly to the user without disturbing others. They are commonly used for personal listening or communication.

PAPER 5

4. a. The storage sizes of flash disks, such as 2GB, 4GB, etc., typically follow a binary numbering system. In binary, each digit can have two possible values: 0 or 1. This binary system is used in digital electronics because it aligns with the underlying hardware architecture, which is based on binary logic gates.

The storage capacities of flash disks are determined by the number of binary digits, or bits, used to represent the data. In the case of storage sizes like 2GB, 4GB, 8GB, etc., the capacities are typically expressed in powers of two. This is because computers use binary addressing, and powers of two are convenient for representing memory sizes.

When it comes to storage sizes, powers of two provide a natural way to increase capacity by adding more bits. For example, to go from 2GB to 4GB, one additional bit is added to each storage unit. This pattern continues, allowing for easier scalability and efficient use of memory addressing in computer systems.

On the manufacturing side, flash memory is typically produced in powers of two capacities due to the way memory cells are organized and addressed. It simplifies the manufacturing process and makes it easier to ensure uniformity and

compatibility across different devices.

b. A 32-bit computer system can support a maximum of 4GB of RAM due to its memory addressing limitations. In a 32-bit architecture, memory addresses are represented using 32 bits. Each bit can represent a binary value of 0 or 1, allowing for a total of 2^32 unique memory addresses.

The memory address space is divided into different regions, including the RAM (Random Access Memory) used for temporary storage while the computer is running. With 32 bits, the addressable memory space is limited to 2^32 addresses, which is equivalent to 4GB (2^32 bytes).

The limitation arises because each memory address needs to be uniquely identified by a binary number. If the system were to support more than 4GB of RAM, it would require more memory addresses, which would exceed the capacity of a 32-bit architecture.

To overcome this limitation, modern computer systems have transitioned to 64-bit architectures, which use 64-bit memory addresses. A 64-bit system can theoretically support up to 18.4 million terabytes of RAM, providing significantly more memory capacity compared to a 32-bit system.

1. Definitions:

a. Binary Number System: The binary number system is a base-2 numeral system used in digital systems, computers, and other electronic devices. It uses only two symbols, typically represented as 0 and 1, to express any numerical value. In this system, each digit's position represents a power of two, allowing for efficient representation and manipulation of binary data.

b. ASCII: ASCII stands for the American Standard Code for Information Interchange. It is a character encoding standard that assigns unique numeric codes to represent characters such as letters, digits,

punctuation marks, and control characters in computers and other devices. ASCII uses 7 bits to represent 128 characters, including the English alphabet, numbers, and common symbols.

c. UNICODE: UNICODE is a character encoding standard that aims to support all

characters from all writing systems used in the world. It provides a unique numerical code, called a code point, for each character, including symbols, letters, and ideograms. Unlike ASCII, UNICODE uses a variable-length encoding scheme, allowing for representation of a much larger range of characters.

d. Network topologies: Network topologies refer to the physical or logical layout of interconnected devices in a computer network. It defines how nodes, such as computers, servers, routers, and switches, are connected and how data flows between them. Common network topologies include

bus, star, ring, mesh, and tree topologies, each with its own advantages and disadvantages.

e. Multiplexing: Multiplexing is a technique used to combine multiple signals or data streams into a single transmission medium or communication channel. It allows for efficient use of resources by transmitting multiple signals simultaneously, either by dividing the available bandwidth into smaller channels (frequency division multiplexing) or by interleaving the signals in time (time division multiplexing).

2. Differentiations:

a. CPU and System Unit: The CPU (Central Processing Unit) is the core component of a computer responsible for executing instructions and performing calculations. It consists of the arithmetic logic unit (ALU), control unit, and registers. The system unit, on the other hand, refers to the main enclosure that houses various components of a computer, including the CPU, memory, storage devices, and other peripherals.

b. RAM and ROM: RAM (Random Access Memory) and ROM (Read-Only Memory) are both types of computer memory, but they have different characteristics. RAM is a volatile memory that provides temporary storage for data and program instructions that are actively being used by the CPU. It allows for fast read and write operations but loses its content when power is turned off. ROM, on the other hand, is non-volatile memory that contains pre-programmed data or instructions that cannot be modified by normal computer operations. It retains its content even when power is removed.

c. Transistor and IC: A transistor is a semiconductor device that amplifies or switches electronic signals and forms the fundamental building block of electronic circuits. It has three layers of semiconductor material and can act as a switch or an amplifier depending on its configuration. An IC (Integrated Circuit) is a miniaturized electronic circuit that consists of multiple transistors, resistors, capacitors, and other components fabricated on a small silicon chip. ICs are used to create complex electronic systems and provide compactness and improved performance compared to discrete components.

d. Synchronous and Asynchronous signal transmission: Synchronous transmission refers to a communication method where data is transmitted in a synchronized manner using a common clock signal. The sending and receiving devices are synchronized, and data is sent in fixed time intervals or

with explicit timing signals. Asynchronous transmission, on the other hand, does not rely on a common clock signal. Each data unit is accompanied by start and stop bits to indicate the beginning and end of a transmission. Asynchronous transmission is commonly used for transmitting data in a character-by-character manner.

e. Full Duplex and Half Duplex: Full duplex and half duplex are modes of communication that determine whether data can be transmitted in both directions simultaneously or not. In full duplex communication, data can be transmitted simultaneously in both directions, allowing for bidirectional communication at the same time. In contrast, in half duplex communication, data can be transmitted in both directions, but not simultaneously. The communication channel alternates between transmitting and receiving data, allowing bidirectional communication but not simultaneously in both directions.

3. Generations of Computers since 1950:

a. First Generation (1950s-1960s): The first generation of computers used vacuum tubes for circuitry and magnetic drums for memory. These machines were large, expensive, and had limited computational capabilities. Examples include the UNIVAC and ENIAC computers.

b. Second Generation (1960s-1970s): The second generation saw the introduction

of transistors, which replaced vacuum tubes, resulting in smaller, more reliable, and more energy-efficient computers. Magnetic core memory was used for storage. Examples include the IBM 1401 and IBM 7094.

c. Third Generation (1960s-1970s): The third generation introduced integrated circuits (ICs), which allowed for even greater miniaturization and increased computational power. Key developments included the use of smaller transistors and the development of operating systems. Examples include the IBM System/360 and DEC PDP-10.

d. Fourth Generation (1970s-1980s): The fourth generation was characterized by the widespread use of microprocessors, which combined multiple integrated circuits on a single chip. This led to the development of personal computers and made computing more accessible. Examples include the Apple II and IBM PC.

e. Fifth Generation (1980s-Present): The fifth generation focused on artificial intelligence and parallel processing. Advanced technologies such as supercomputers, expert systems, and neural networks were developed. The emphasis was on creating machines that could mimic human intelligence and solve complex problems. This generation is ongoing, with advancements in areas such as machine learning and quantum computing.

5. Types of Networks and Network Topologies:

Types of Networks:

a. Local Area Network (LAN): A LAN is a network that covers a small geographical area, such as a home, office, or campus. It connects computers and devices in close proximity, allowing for high-speed data transfer and resource sharing.

b. Wide Area Network (WAN): A WAN spans a large geographical area, such as multiple offices or cities. It connects LANs and other networks over long distances using various communication technologies, such as leased lines, satellites, or internet connections.

c. Metropolitan Area Network (MAN): A MAN covers a city or metropolitan area, providing connectivity between multiple LANs and institutions within the same

region. It typically uses high-speed fiber optic or wireless connections.

d. Wireless Network: A wireless network uses wireless communication technologies, such as Wi-Fi, to connect devices without the need for physical cables. It allows for flexible and mobile connectivity within a specific range.

Network Topologies:

a. Bus Topology: In a bus topology, all devices are connected to a central cable called a bus. Data is transmitted along the bus, and each device receives the transmitted data. It is a simple and cost-effective topology but can be prone to network congestion and single point of failure.

b. Star Topology: In a star topology, all devices are connected to a central device, such as a switch or hub. Data is transmitted through the central device, which manages the communication between devices. It provides better performance and scalability but relies on the central device's availability.

c. Ring Topology: In a ring topology, devices are connected in a closed loop, where each device is connected to two neighboring devices. Data circulates around the ring, and each device receives and transmits data sequentially. It provides equal access to all devices but can be vulnerable to a single point of failure.

d. Mesh Topology: In a mesh topology, each device is connected to every other device in the network, creating multiple paths for data transmission. It offers high reliability and fault tolerance but requires more cabling and can be complex to manage.

e. Tree Topology: A tree topology combines multiple star topologies in a hierarchical structure. Devices are connected to central devices, which are then connected to other central devices, forming a tree-like structure. It allows for scalability and efficient management of large networks.

PAPER 6

Question One:

a) The statement contains incorrect information. The first computers did not

use integrated circuits (IC) for their circuitry; they used vacuum tubes. Integrated circuits were developed later. Additionally, magnetic drums were used for primary memory, not secondary memory. These early computers were not very portable and were housed in large rooms or buildings. They were also not very efficient in the use of electricity and generated a significant amount of heat, which often caused malfunctions.

b) Definitions:

a. Program: A program is a set of instructions or commands written in a programming language that tells a computer what tasks to perform. It is a sequence of instructions that the computer follows to execute a specific task or achieve a desired outcome.

b. Command: A command is a specific instruction given to a computer system or program to perform a particular action. Commands can be issued through various means, such as a command-line interface (CLI) or a graphical user interface (GUI), and they allow users to interact with the computer and initiate specific operations.

c) The four sizes of computers from the largest to the smallest are:

1. Supercomputers: These are the largest and most powerful computers, designed to process massive amounts of data and perform complex calculations. They are used for scientific research, weather forecasting, and simulations.

2. Mainframe Computers: Mainframes are large computers used by organizations for data processing, handling large-scale business applications, and running multiple concurrent tasks or users.

3. Minicomputers: Minicomputers are smaller than mainframes but larger than microcomputers. They are capable of supporting multiple users simultaneously and are used in small to medium-sized businesses.

4. Microcomputers/Personal Computers: Microcomputers, commonly known as personal computers (PCs), are the smallest computers. They are designed for individual use and can perform a wide range of tasks such as word processing, internet browsing, and gaming.

The major functions of each size of computer are:

- Supercomputers: High-performance computing, complex simulations, scientific research.

- Mainframe Computers: Large-scale data processing, transaction processing, centralized storage and management of data.

- Minicomputers: Departmental computing, network servers, mid-range business applications.

- Microcomputers/Personal Computers: Individual productivity, gaming, internet browsing, personal tasks.

d) Two factors that make computers powerful are:

1. Processing Power: The speed and capabilities of the computer's central processing unit (CPU)

significantly impact its power. A faster CPU allows for quicker execution of instructions and handling complex calculations.

2. Memory Capacity: The amount of random access memory (RAM) in a computer affects its ability to handle and store data. More RAM enables the computer to run multiple programs simultaneously and handle larger datasets efficiently.